Package ‘mcglm’

November 27, 2025

Type Package

Title Multivariate Covariance Generalized Linear Models

Version 0.9.0

Date 2025-11-27

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Description Fitting multivariate covariance generalized linear models (McGLMs) to data. McGLM is a general framework for non-normal multivariate data analysis, designed to handle multivariate response variables, along with a wide range of temporal and spatial correlation structures defined in terms of a covariance link function combined with a matrix linear predictor involving known matrices.
The models take non-normality into account in the conventional way by means of a variance function, and the mean structure is modelled by means of a link function and a linear predictor.
The models are fitted using an efficient Newton scoring algorithm based on quasi-likelihood and Pearson estimating functions, using only second-moment assumptions. This provides a unified approach to a wide variety of different types of response variables and covariance structures, including multivariate extensions of repeated measures, time series, longitudinal, spatial and spatio-temporal structures.
The package offers a user-friendly interface for fitting McGLMs similar to the glm() R function.
See Bonat (2018) <doi:10.18637/jss.v084.i04>, for more information and examples.

Depends R (>= 4.5.0)

Suggests testthat, knitr, rmarkdown, MASS, mvtnorm, tweedie, devtools

Imports stats, Matrix, assertthat, graphics, Rcpp (>= 0.12.16)

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LazyData TRUE

URL mcglm.leg.ufpr.br

BugReports https://github.com/wbonat/mcglm/issues

Encoding UTF-8

VignetteBuilder knitr

RoxygenNote 7.3.3

LinkingTo Rcpp, RcppArmadillo

NeedsCompilation yes

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`ahs` *Australian Health Survey*

Description

The Australian health survey was used by Bonat and Jorgensen (2016) as an example of multivariate count regression model. The data consists of five count response variables concerning health system access measures and nine covariates concerning social conditions in Australian for 1987-88.

- `sex` - Factor with levels `male` and `female`.
- `age` - Respondent's age in years divided by 100.
- `income` - Respondent's annual income in Australian dollars divided by 1000.
- `levyplus` - Coded factor. If respondent is covered by private health insurance fund for private patients in public hospital with doctor of choice (1) or otherwise (0).
- `freepoor` - Coded factor. If respondent is covered by government because low income, recent immigrant, unemployed (1) or otherwise (0).
- `freerepa` - Coded factor. If respondent is covered free by government because of old-age or disability pension, or because invalid veteran or family of deceased veteran (1) or otherwise (0).

- illnes - Number of illnesses in past 2 weeks, with 5 or illnesses coded as 5.
- actdays - Number of days of reduced activity in the past two weeks due to illness or injury.
- hscore - Respondent's general health questionnaire score using Goldberg's method. High score indicates poor health.
- chcond - Factor with three levels. If respondent has chronic condition(s) and is limited in activity (limited), or if the respondent has chronic condition(s) but is not limited in activity (nonlimited) or otherwise (otherwise, reference level).
- Ndoc - Number of consultations with a doctor or specialist (response variable).
- Nndoc - Number of consultations with health professionals (response variable).
- Nadm - Number of admissions to a hospital, psychiatric hospital, nursing or convalescence home in the past 12 months (response variable).
- Nhosp - Number of nights in a hospital during the most recent admission.
- Nmed - Total number of prescribed and non prescribed medications used in the past two days.

Usage

```
data(ahs)
```

Format

a data.frame with 5190 records and 15 variables.

Source

Deb, P. and Trivedi, P. K. (1997) Demand for medical care by the elderly: A finite mixture approach. Journal of Applied Econometrics 12(3):313–336.

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

Examples

```
require(mcglm)
data(ahs, package="mcglm")
form1 <- Ndoc ~ income + age
form2 <- Nndoc ~ income + age
Z0 <- mc_id(ahs)
fit.ahs <- mcglm(linear_pred = c(form1, form2),
                 matrix_pred = list(Z0, Z0), link = c("log","log"),
                 variance = c("poisson_tweedie","poisson_tweedie"),
                 data = ahs)
summary(fit.ahs)
```

`anova.mcglm` *Anova Tables*

Description

Performs Wald tests of the significance for the linear predictor components by response variables. This function is useful for joint hypothesis tests of regression coefficients associated with categorical covariates with more than two levels. It is not designed for model comparison.

Usage

```
## S3 method for class 'mcglm'
anova(object, ...)
```

Arguments

`object`	an object of class `mcglm`, usually, a result of a call to `mcglm()` function.
`...`	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

A `data.frame` with Chi-square statistic to test the null hypothesis of a parameter, or a set of parameters, be zero. Degree of freedom (Df) and p-values. The Wald test based on the observed covariance matrix of the parameters is used.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Examples

```
x1 <- seq(-1, 1, l = 100)
x2 <- gl(5, 20)
beta <- c(5, 0, -2, -1, 1, 2)
X <- model.matrix(~ x1 + x2)
set.seed(123)
y <- rnorm(100, mean = X%*%beta, sd = 1)
data = data.frame("y" = y, "x1" = x1, "x2" = x2)
fit.anova <- mcglm(c(y ~ x1 + x2), list(mc_id(data)), data = data)
anova(fit.anova)
```

coef.mcglm *Model Coefficients*

Description

Extract model coefficients for objects of mcglm class.

Usage

```
## S3 method for class 'mcglm'
coef(
  object,
  std.error = FALSE,
  response = c(NA, 1:length(object$beta_names)),
  type = c("beta", "tau", "power", "correlation"),
  ...
)
```

Arguments

object	an object of mcglm class.
std.error	logical. If TRUE returns the standard errors for the estimates. Default is FALSE.
response	a numeric vector specifyng for which response variable the coefficients should be returned.
type	a string vector (can be 1 element length) specifying which coefficients should be returned. Options are "beta", "tau", "power", "tau" and "correlation".
...	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

A data.frame with parameters names, estimates, response variable number and parameters type.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

confint.mcglm *Confidence Intervals for Model Parameters*

Description

Computes confidence intervals for parameters in a fitted mcglm model.

Usage

```
## S3 method for class 'mcglm'
confint(object, parm, level = 0.95, ...)
```

Arguments

`object`	a fitted `mcglm` object.
`parm`	specifies for which parameters are to be given confidence intervals, either a vector of numbers or a vector of strings. If missing, all parameters are considered.
`level`	the nominal confidence level.
`...`	additional arguments affecting the confidence interval produced. Note that there is no extra options for `mcglm` object class.

Value

A `data.frame` with confidence intervals, parameters names, response variable number and parameters type.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

---

covprod *Cross variability matrix*

Description

Compute the cross-covariance matrix between covariance and regression parameters. Equation (11) of Bonat and Jorgensen (2016).

Usage

```
covprod(A, res, W)
```

Arguments

`A`	A matrix.
`res`	A vector of residuals.
`W`	A matrix of weights.

Author(s)

Wagner Hugo Bonat

`ESS` *Generalized Error Sum of Squares*

Description

Extract the generalized error sum of squares (ESS) for objects of `mcglm` class.

Usage

```
ESS(object, verbose = TRUE)
```

Arguments

`object`	an object or a list of objects representing a model of `mcglm` class.
`verbose`	logical. Print or not the ESS value.

Value

Returns the value of the generalized error sum of squares (ESS).

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Wang, M. (2014). Generalized Estimating Equations in Longitudinal Data Analysis: A Review and Recent Developments. Advances in Statistics, 1(1)1–13.

See Also

`gof`, `plogLik`, `pAIC`, `pKLIC`, `GOSHO` and `RJC`.

`fitted.mcglm` *Fitted Values*

Description

Extract fitted values for objects of `mcglm` class.

Usage

```
## S3 method for class 'mcglm'
fitted(object, ...)
```

Arguments

object	an object of mcglm class.
...	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

A matrix with fitted values.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

---

fit_mcglm *Chaser and Reciprocal Likelihood Algorithms*

Description

This function implements the two main algorithms used for fitting multivariate covariance generalized linear models. The chaser and the reciprocal likelihood algorithms.

Usage

```
fit_mcglm(list_initial, list_link, list_variance,
         list_covariance, list_X, list_Z, list_offset,
         list_Ntrial, list_power_fixed, list_sparse,
         y_vec, correct, max_iter, tol, method,
         tuning, verbose, weights)
```

Arguments

list_initial	a list of initial values for regression and covariance parameters.
list_link	a list specifying the link function names. Options are: "logit", "probit", "cauchit", "cloglog", "loglog", "identity", "log", "sqrt", "1/mu^2" and "inverse". See mc_link_function for details. Default link = "identity".
list_variance	a list specifying the variance function names. Options are: "constant", "tweedie", "poisson_tweedie", "binomialP" and "binomialPQ". See mc_variance_function for details. Default variance = "constant".
list_covariance	a list of covariance function names. Options are: "identity", "inverse" and "expm". Default covariance = "identity".
list_X	a list of design matrices. See model.matrix for details.
list_Z	a list of knowm matrices to compose the matrix linear predictor.
list_offset	a list of offset values. Default NULL.
list_Ntrial	a list of number of trials, useful only when analysing binomial data. Default 1.
list_power_fixed	a list of logicals indicating if the power parameters should be estimated or not. Default power_fixed = TRUE.

list_sparse	a list of logicals indicating if the matrices should be set up as sparse matrices. This argument is useful only when using exponential-matrix covariance link function. In the other cases the algorithm detects automatically if the matrix should be sparse or not.
y_vec	a vector of the stacked response variables.
correct	a logical indicating if the algorithm will use the correction term or not. Default correct = FALSE.
max_iter	maximum number of iterations. Default max_iter = 20.
tol	a numeric specyfing the tolerance. Default tol = 1e-04.
method	a string specyfing the method used to fit the models ("chaser" or "rc"). Default method = "chaser".
tuning	a numeric value in general close to zero for the rc method and close to 1 for the chaser method. This argument control the step-length. Default tuning = 1.
verbose	a logical if TRUE print the values of the covariance parameters used on each iteration. Default verbose = FALSE
weights	Vector of weights for model fitting.

Value

A list with regression and covariance parameter estimates. Details about the estimation procedures as iterations, sensitivity, variability are also provided. In general the users do not need to use this function directly. The mcglm provides GLM interface for fitting mcglm.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mcglm, mc_matrix_linear_predictor, mc_link_function and mc_variance_function.

---

gof *Measures of Goodness-of-Fit*

Description

Extract the pseudo Gaussian log-likelihood (plogLik), pseudo Akaike Information Criterion (pAIC), pseudo Kullback-Leibler Information Criterion (pKLIC) and pseudo Bayesian Information Criterion (pBIC) for objects of mcglm class.

Usage

```
gof(object)
```

Arguments

object	an object or a list of objects representing a model of mcglm class.

Value

Returns a data frame containing goodness-of-fit measures.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Wang, M. (2014). Generalized Estimating Equations in Longitudinal Data Analysis: A Review and Recent Developments. Advances in Statistics, 1(1)1–13.

See Also

plogLik, pAIC, pKLIC and pBIC.

GOSHO *Gosho Information Criterion*

Description

Extract the Gosho Information Criterion (GOSHO) for an object of mcglm class. WARNING: This function is limited to models with ONE response variable. This function is general useful only for longitudinal data analysis.

Usage

```
GOSHO(object, id, verbose = TRUE)
```

Arguments

object	an object of mcglm class.
id	a vector which identifies the clusters or groups. The length and order of id should be the same as the number of observations. Data are assumed to be sorted so that observations on a cluster are contiguous rows for all entities in the formula.
verbose	logical. Print or not the GOSHO value.

Value

The value of the GOSHO criterion. Note that the function assumes that the data are in the correct order.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Wang, M. (2014). Generalized Estimating Equations in Longitudinal Data Analysis: A Review and Recent Developments. Advances in Statistics, 1(1)1–13.

See Also

`gof`, `plogLik`, `pAIC`, `pKLIC`, `ESS` and `RJC`.

`Hunting` *Hunting in Pico Basile, Bioko Island, Equatorial Guinea.*

Description

Case study analysed in Bonat et. al. (2016) concernings on data of animals hunted in the village of Basile Fang, Bioko Norte Province, Bioko Island, Equatorial Guinea. Monthly number of blue duikers and other small animals shot or snared was collected for a random sample of 52 commercial hunters from August 2010 to September 2013. For each animal caught, the species, sex, method of capture and altitude were documented. The data set has 1216 observations.

- `ALT` - Factor five levels indicating the Altitude where the animal was caught.
- `SEX` - Factor two levels `Female` and `Male`.
- `METHOD` - Factor two levels `Escopeta` and `Trampa`.
- `OT` - Monthly number of other small animals hunted.
- `BD` - Monthly number of blue duikers hunted.
- `OFFSET` - Monthly number of hunter days.
- `HUNTER` - Hunter index.
- `MONTH` - Month index.
- `MONTHCALENDAR` - Month using calendar numbers (1-January, ..., 12-December).
- `YEAR` - Year calendar (2010–2013).
- `HUNTER.MONTH` - Index indicating observations taken at the same HUNTER and MONTH.

Usage

```
data(Hunting)
```

Format

a `data.frame` with 1216 records and 11 variables.

Source

Bonat, et. al. (2017). Modelling the covariance structure in marginal multivariate count models: Hunting in Bioko Island. Journal of Agricultural Biological and Environmental Statistics, 22(4):446–464.

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Examples

```
library(mcglm)
library(Matrix)
data(Hunting, package="mcglm")
formu <- OT ~ METHOD*ALT + SEX + ALT*poly(MONTH, 4)
Z0 <- mc_id(Hunting)
Z1 <- mc_mixed(~0 + HUNTER.MONTH, data = Hunting)
fit <- mcglm(linear_pred = c(formu), matrix_pred = list(c(Z0, Z1)),
             link = c("log"), variance = c("poisson_tweedie"),
             power_fixed = c(FALSE),
             control_algorithm = list(max_iter = 100),
             offset = list(log(Hunting$OFFSET)), data = Hunting)
summary(fit)
anova(fit)
```

---

mcglm — *Fitting Multivariate Covariance Generalized Linear Models*

Description

The function `mcglm` is used to fit multivariate covariance generalized linear models. The models are specified by a set of lists giving a symbolic description of the linear and matrix linear predictors. The user can choose between a list of link, variance and covariance functions. The models are fitted using an estimating function approach, combining quasi-score functions for regression parameters and Pearson estimating function for covariance parameters. For details see Bonat and Jorgensen (2016).

Usage

```
mcglm(linear_pred, matrix_pred, link, variance, covariance,
       offset, Ntrial, power_fixed, data, control_initial,
       contrasts, weights, control_algorithm)
```

Arguments

`linear_pred`	a list of formula see `formula` for details.
`matrix_pred`	a list of known matrices to be used on the matrix linear predictor. For details see `mc_matrix_linear_predictor`.
`link`	a list of link functions names. Options are: `"logit"`, `"probit"`, `"cauchit"`, `"cloglog"`, `"loglog"`, `"identity"`, `"log"`, `"sqrt"`, `"1/mu^2"` and `"inverse"`. See `mc_link_function` for details.
`variance`	a list of variance functions names. Options are: `"constant"`, `"tweedie"`, `"poisson_tweedie"`, `"binomialP"` and `"binomialPQ"`. See `mc_variance_function` for details.
`covariance`	a list of covariance link functions names. Options are: `"identity"`, `"inverse"` and exponential-matrix `"expm"`.
`offset`	a list of offset values if any.
`Ntrial`	a list of number of trials on Bernoulli experiments. It is useful only for `binomialP` and `binomialPQ` variance functions.

power_fixed	a list of logicals indicating if the values of the power parameter should be estimated or not.
data	a data frame.
control_initial	a list of initial values for the fitting algorithm. If no values are supplied automatic initial values will be provided by the function mc_initial_values.
contrasts	extra arguments to passed to model.matrix.
weights	A list of weights for model fitting. Each element of the list should be a vector of weights of size equals the number of observations. Missing observations should be annotated as NA.
control_algorithm	a list of arguments to be passed for the fitting algorithm. See fit_mcglm for details.

Value

mcglm returns an object of class 'mcglm'.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

fit_mcglm, mc_link_function and mc_variance_function.

---

mc_anova_disp *Anova Tables for dispersion components*

Description

IT IS AN EXPERIMENTAL FUNCTION BE CAREFUL! Performs Wald tests of the significance for the dispersion components by response variables. This function is useful for joint hypothesis tests of dispersion coefficients associated with categorical covariates with more than two levels. It is not designed for model comparison.

Usage

```
mc_anova_disp(object, idx_list, names_list, ...)
```

Arguments

`object`	an object of class `mcglm`, usually, a result of a call to `mcglm()` function.
`idx_list`	list with indexes for parameter tests.
`names_list`	list of names to appear in the anova table.
`...`	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

A `data.frame` with Chi-square statistic to test the null hypothesis of a parameter, or a set of parameters, be zero. Degree of freedom (Df) and p-values. The Wald test based on the observed covariance matrix of the parameters is used.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Examples

```
x1 <- seq(0, 1, l = 100)
x2 <- gl(5, 20)
beta <- c(5, 0, -2, -1, 1, 2)
X <- model.matrix(~ x1 + x2)
set.seed(123)
y <- rnorm(100, mean = 10, sd = X%*%beta)
data = data.frame("y" = y, "x1" = x1, "x2" = x2, "id" = 1)
fit.anova <- mcglm(c(y ~ 1), list(mc_dglm(~ x1 + x2, id = "id", data)),
                   control_algorithm = list(tuning = 0.9), data = data)
X <- model.matrix(~ x1 + x2, data = data)
idx <- attr(X, "assign")
idx_list <- list("idx" = idx, "idx" = idx)
names_list <- list(colnames(X), colnames(X))
mc_anova_disp(object = fit.anova, idx = idx_list, names_list = names_list)
```

`mc_bias_corrected_std` *Bias-corrected Standard Error for Regression Parameters*

Description

Compute bias-corrected standard error for regression parameters in the context of clustered observations for an object of `mcglm` class. It is also robust and has improved finite sample properties.

Usage

```
mc_bias_corrected_std(object, id)
```

Arguments

`object`	an object of `mcglm` class.
`id`	a vector which identifies the clusters. The length and order of `id` should be the same as the number of observations. The data set are assumed to be sorted so that observations on a cluster are contiguous rows for all entities.

Value

A variance-covariance matrix. Note that the function assumes that the data are in the correct order.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Nuamah, I. F. and Qu, Y. and Aminu, S. B. (1996). A SAS macro for stepwise correlated binary regression. Computer Methods and Programs in Biomedicine 49, 199–210.

See Also

`mc_robust_std`.

`mc_build_bdiag` *Build a block-diagonal matrix of zeros.*

Description

Build a block-diagonal matrix of zeros. Such functions is used when computing the derivatives of the Cholesky decomposition of C.

Usage

```
mc_build_bdiag(n_resp, n_obs)
```

Arguments

`n_resp`	A numeric specifyng the number of response variables.
`n_obs`	A numeric specifying the number of observations in the data set.

Details

It is an internal function.

Value

A list of zero matrices.

Author(s)

Wagner Hugo Bonat

mc_build_C *Build the joint covariance matrix*

Description

This function builds the joint variance-covariance matrix using the Generalized Kronecker product and its derivatives with respect to rho, power and tau parameters.

Usage

```
mc_build_C(
  list_mu,
  list_Ntrial,
  rho,
  list_tau,
  list_power,
  list_Z,
  list_sparse,
  list_variance,
  list_covariance,
  list_power_fixed,
  compute_C = FALSE,
  compute_derivative_beta = FALSE,
  compute_derivative_cov = TRUE
)
```

Arguments

list_mu	A list with values of the mean.
list_Ntrial	A list with the number of trials. Usefull only for binomial responses.
rho	Vector of correlation parameters.
list_tau	A list with values for the tau parameters.
list_power	A list with values for the power parameters.
list_Z	A list of matrix to be used in the matrix linear predictor.
list_sparse	A list with Logical.
list_variance	A list specifying the variance function to be used for each response variable.
list_covariance	A list specifying the covariance function to be used for each response variable.
list_power_fixed	A list of Logical specifying if the power parameters are fixed or not.
compute_C	Logical. Compute or not the C matrix.
compute_derivative_beta	Logical. Compute or not the derivative of C with respect to regression parameters.
compute_derivative_cov	Logical. Compute or not the derivative of C with respect the covariance parameters.

Value

A list with the inverse of the C matrix and the derivatives of the C matrix with respect to rho, power and tau parameters.

Author(s)

Wagner Hugo Bonat

`mc_build_F` *Auxiliar function: Build F matrix for Wald multivariate test*

Description

The function `mc_build_F` is just an auxiliar function to construct the design matrix for multivariate Wald-type test.

Usage

```
mc_build_F(vector)
```

Arguments

`vector`	A vector indexing model regression coefficients.

Value

A matrix.

Author(s)

Wagner Hugo Bonat

`mc_build_omega` *Build omega matrix*

Description

This function build Ω matrix according the covariance link function.

Usage

```
mc_build_omega(tau, Z, covariance_link, sparse = FALSE)
```

Arguments

`tau`	A vector
`Z`	A list of matrices.
`covariance_link`	String specifing the covariance link function: identity, inverse, expm.
`sparse`	Logical force to use sparse matrix representation 'dsCMatrix'.

Value

A list with the Ω matrix its inverse and derivatives with respect to τ.

Author(s)

Wagner Hugo Bonat

---

mc_build_sigma *Build variance-covariance matrix*

Description

This function builds a variance-covariance matrix, based on the variance function and omega matrix.

Usage

```
mc_build_sigma(
  mu,
  Ntrial = 1,
  tau,
  power,
  Z,
  sparse,
  variance,
  covariance,
  power_fixed,
  compute_derivative_beta = FALSE
)
```

Arguments

`mu`	A numeric vector. In general the output from `mc_link_function`.
`Ntrial`	A numeric vector, or NULL or a numeric specifing the number of trials in the binomial experiment. It is usefull only when using variance = binomialP or binomialPQ. In the other cases it will be ignored.
`tau`	A numeric vector.
`power`	A numeric or numeric vector. It should be one number for all variance functions except binomialPQ, in that case the argument specifies both p and q.
`Z`	A list of matrices.
`sparse`	Logical.
`variance`	String specifing the variance function: constant, tweedie, poisson_tweedie, binomialP or binomialPQ.
`covariance`	String specifing the covariance function: identity, inverse or expm.
`power_fixed`	Logical if the power parameter is fixed at initial value (TRUE). In the case power_fixed = FALSE the power parameter will be estimated.
`compute_derivative_beta`	Logical. Compute or not the derivative with respect to regression parameters.

Value

A list with the Cholesky decomposition of Σ, Σ^{-1} and the derivative of Σ with respect to the power and tau parameters.

Author(s)

Wagner Hugo Bonat

See Also

mc_link_function, mc_variance_function, mc_build_omega.

---

mc_build_sigma_between *Build the correlation matrix between response variables*

Description

This function builds the correlation matrix between response variable, its inverse and derivatives.

Usage

```
mc_build_sigma_between(rho, n_resp, inverse = FALSE)

mc_derivative_sigma_between(n_resp)
```

Arguments

rho	A numeric vector.
n_resp	A numeric.
inverse	Logical.

Value

A list with sigmab and its derivatives with respect to rho.

Author(s)

Wagner Hugo Bonat

mc_car *Conditional Autoregressive Model Structure*

Description

The function mc_car helps to build the components of the matrix linear predictor used for fitting conditional autoregressive models. This function is used in general for fitting spatial areal data using the well known conditional autoregressive models (CAR). This function depends on a list of neighboors, such a list can be constructed, for example using the tri2nb function from the spdep package based on spatial coordinates. This way to specify the matrix linear predictor can also be applied for spatial continuous data, as an approximation.

Usage

```
mc_car(list_neigh, intrinsic = FALSE)
```

Arguments

list_neigh	list of neighboors.
intrinsic	logical.

Value

A list of a matrix (intrinsic = TRUE) or two matrices (intrinsic = FALSE).

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_id, mc_compute_rho, mc_conditional_test, mc_dist, mc_ma, mc_rw
and mc_mixed.

mc_complete_data *Complete Data Set with NA*

Description

The function mc_complete_data completes a data set with NA values for helping to construct the components of the matrix linear predictor in models that require equal number of observations by subjects (id).

Usage

```
mc_complete_data(data, id, index, id.exp)
```

Arguments

data	a data.frame to be completed with NA.
id	name of the column (string) containing the subject id.
index	name of the column (string) containing the index to be completed.
id.exp	how the index is expected to be for all subjects.

Value

A data.frame with the same number of observations by subject.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_dglm, mc_ns, mc_ma and mc_rw.

---

mc_compute_rho *Autocorrelation Estimates*

Description

Compute autocorrelation estimates based on a fitted model using the mc_car structure. The mcglm approach fits models using a linear covariance structure, but in general in this parametrization for spatial models the parameters have no simple interpretation in terms of spatial autocorrelation. The function mc_compute_rho computes the autocorrelation based on a fitted model.

Usage

```
mc_compute_rho(object, level = 0.975)
```

Arguments

object	an object or a list of objects representing a model of mcglm class.
level	the confidence level required.

Value

Returns estimate, standard error and confidential interval for the spatial autocorrelation parameter.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`mc_car` and `mc_conditional_test`.

`mc_conditional_test` *Conditional Hypotheses Tests*

Description

Compute conditional hypotheses tests for fitted `mcglm` model class. When fitting models with extra power parameters, the standard errors associated with the dispersion parameters can be large. In that cases, we suggest to conduct conditional hypotheses test instead of the orthodox marginal test for the dispersion parameters. The function `mc_conditional_test` offers an ease way to conduct such conditional test. Furthermore, the function is quite flexible and can be used for any other conditional hypotheses test.

Usage

```
mc_conditional_test(object, parameters, test, fixed)
```

Arguments

`object`	an object representing a model of `mcglm` class.
`parameters`	which parameters will be included in the conditional test.
`test`	index indicating which parameters will be tested given the values of the other parameters.
`fixed`	index indicating which parameters should be fixed on the conditional test.

Value

Returns estimates, conditional standard errors and Z-statistics.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

mc_core_pearson *Core of the Pearson estimating function.*

Description

Core of the Pearson estimating function.

Usage

```
mc_core_pearson(product, inv_C, res, W)
```

Arguments

product	A matrix.
inv_C	A matrix.
res	A vector of residuals.
W	Matrix of weights.

Details

It is an internal function.

Value

A vector

Author(s)

Wagner Hugo Bonat

mc_correction *Pearson correction term*

Description

Compute the correction term associated with the Pearson estimating function.

Usage

```
mc_correction(D_C, inv_J_beta, D, inv_C)
```

Arguments

D_C	A list of matrices.
inv_J_beta	A matrix. In general it is computed based on the output of the `[mcglm]{mc_quasi_score}`.
D	A matrix. In general it is the output of the mc_link_function.
inv_C	A matrix. In general the output of the mc_build_C.

Details

It is an internal function useful inside the fitting algorithm.

Value

A vector with the correction terms to be used on the Pearson estimating function.

Author(s)

Wagner Hugo Bonat

`mc_cross_sensitivity` *Cross-sensitivity*

Description

Compute the cross-sensitivity matrix between regression and covariance parameters. Equation 10 of Bonat and Jorgensen (2015).

Usage

```
mc_cross_sensitivity(
  Product_cov,
  Product_beta,
  n_beta_effective = length(Product_beta)
)
```

Arguments

`Product_cov`	A list of matrices.
`Product_beta`	A list of matrices.
`n_beta_effective`	Numeric. Effective number of regression parameters.

Value

The cross-sensitivity matrix. Equation (10) of Bonat and Jorgensen (2016).

Author(s)

Wagner Hugo Bonat

mc_cross_variability *Compute the cross-variability matrix*

Description

Compute the cross-variability matrix between covariance and regression parameters.

Usage

```
mc_cross_variability(Product_cov, inv_C, res, D)
```

Arguments

Product_cov	A list of matrices.
inv_C	A matrix.
res	A vector.
D	A matrix.

Value

The cross-variability matrix between regression and covariance parameters.

Author(s)

Wagner Hugo Bonat

mc_derivative_cholesky *Derivatives of the Cholesky decomposition*

Description

This function compute the derivative of the Cholesky decomposition.

Usage

```
mc_derivative_cholesky(derivada, inv_chol_Sigma, chol_Sigma)
```

Arguments

derivada	A matrix.
inv_chol_Sigma	A matrix.
chol_Sigma	A matrix.

Details

It is an internal function.

Value

A list of matrix.

Author(s)

Wagner Hugo Bonat

---

mc_derivative_C_rho *Derivative of C with respect to rho.*

Description

Compute the derivative of the C matrix with respect to the correlation parameters rho.

Usage

```
mc_derivative_C_rho(
  D_Sigmab,
  Bdiag_chol_Sigma_within,
  t_Bdiag_chol_Sigma_within,
  II
)
```

Arguments

D_Sigmab	A matrix.
Bdiag_chol_Sigma_within	A block-diagonal matrix.
t_Bdiag_chol_Sigma_within	A block-diagonal matrix.
II	A diagonal matrix.

Details

It is an internal function used to build the derivatives of the C matrix.

Value

A matrix.

Author(s)

Wagner Hugo Bonat

mc_derivative_expm *Derivative of exponential-matrix function*

Description

Compute the derivative of the exponential-matrix covariance link function.

Usage

```
mc_derivative_expm(dU, UU, inv_UU, Q, n = dim(UU)[1], sparse = FALSE)
```

Arguments

`dU`	A matrix.
`UU`	A matrix.
`inv_UU`	A matrix.
`Q`	A numeric vector.
`n`	A numeric.
`sparse`	Logical.

Details

Many arguments required by this function are provide by the `link[mcglm]{mc_dexpm}`. The argument dU is the derivative of the U matrix with respect to the models parameters. It should be computed by the user.

Value

A matrix.

Author(s)

Wagner Hugo Bonat

See Also

`expm`, `link[mcglm]{mc_dexp_gold}` and `link[mcglm]{mc_dexpm}`.

`mc_derivative_sigma_beta` *Derivatives of $V\hat{}1/2$ with respect to beta.*

Description

Compute the derivatives of $V^{1/2}$ matrix with respect to the regression parameters beta.

Usage

```
mc_derivative_sigma_beta(D, D_V_sqrt_mu, Omega, V_sqrt, variance)
```

Arguments

`D`	A matrix.
`D_V_sqrt_mu`	A matrix.
`Omega`	A matrix.
`V_sqrt`	A matrix.
`variance`	A string specifying the variance function name.

Value

A list of matrices, containg the derivatives of $V^{1/2}$ with respect to the regression parameters.

Author(s)

Wagner Hugo Bonat

`mc_dexp_gold` *Exponential-matrix and its derivatives*

Description

Given a matrix M and its derivative dM the function `dexp_gold` returns the exponential-matrix $expm(M)$ and its derivative. This function is based on the `expm` function. It is not really used in the package, but I keep this function to test my own implementation based on eigen values decomposition.

Usage

```
mc_dexp_gold(M, dM)
```

Arguments

`M`	A matrix.
`dM`	A matrix.

Value

A list with two elements: $expm(M)$ and its derivatives.

Author(s)

Wagner Hugo Bonat

See Also

expm, eigen.

Examples

```
M <- matrix(c(1,0.8,0.8,1), 2,2)
dM <- matrix(c(0,1,1,0),2,2)
mcglm::mc_dexp_gold(M = M, dM = dM)
```

---

mc_dglm *Double Generalized Linear Models Structure*

Description

The function `mc_dglm` builds the components of the matrix linear predictor used for fitting double generalized linear models.

Usage

```
mc_dglm(formula, id, data)
```

Arguments

`formula`	a formula spefying the components of the covariance structure.
`id`	name of the column (string) containing the subject index. (If ts is not repeated measures, use id = 1 for all observations).
`data`	data set.

Value

A list of a diagonal matrices, whose values are given by the covariates assumed to describe the covariance structure.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_id, mc_dist, mc_ma, mc_rw
and mc_mixed.

---

mc_dist *Distance Models Structure*

Description

The function mc_dist helps to build the components of the matrix linear predictor using matrices based on distances. This function is generaly used for the analysis of longitudinal and spatial data. The idea is to use the inverse of some measure of distance as for example the Euclidean distance to model the covariance structure within response variables. The model can also use the inverse of distance squared or high order power.

Usage

```
mc_dist(id, time, data, method = "euclidean")
```

Arguments

id	name of the column (string) containing the subject index. For spatial data use the same id for all observations (one unit sample).
time	name of the column (string) containing the index indicating the time. For spatial data use the same index for all observations.
data	data set.
method	distance measure to be used.

Details

The distance measure must be one of "euclidean", "maximum", "manhattan", "canberra", "binary" or "minkowski". This function is a customize call of the dist function.

Value

A matrix of dgCMatrix class.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

dist, mc_id, mc_conditional_test, mc_car, mc_ma, mc_rw and mc_mixed.

Examples

```
id <- rep(1:2, each = 4)
time <- rep(1:4, 2)
data <- data.frame("id" = id, "time" = time)
mc_dist(id = "id", time = "time", data = data)
mc_dist(id = "id", time = "time", data = data, method = "canberra")
```

---

mc_expm — *Exponential-matrix covariance link function*

Description

Given a matrix U the function mc_expm returns the exponential-matrix $expm(U)$ and some auxiliares matrices to compute its derivatives. This function is based on the eigen-value decomposition it means that it is very slow.

Usage

```
mc_expm(U, n = dim(U)[1], sparse = FALSE, inverse = FALSE)
```

Arguments

U	A matrix.
n	A number specifing the dimension of the matrix U. Default n = dim(U)[1].
sparse	Logical defining the class of the output matrix. If sparse = TRUE the output class will be 'dgCMatrix' if sparse = FALSE the class will be 'dgMatrix'.
inverse	Logical defining if the inverse will be computed or not.

Value

A list with $\Omega = expm(U)$ its inverse (if inverse = TRUE) and auxiliares matrices to compute the derivatives.

Author(s)

Wagner Hugo Bonat

See Also

expm, eigen, link[mcglm]{mc_dexp_gold}.

`mc_getInformation` *Getting information about model parameters*

Description

This computes all information required about the number of model parameters.

Usage

```
mc_getInformation(list_initial, list_power_fixed, n_resp)
```

Arguments

`list_initial`	A list of initial values.
`list_power_fixed`	A list of logical specyfing if the power parameters should be estimated or not.
`n_resp`	A number specyfing the nmber of response variables.

Value

The number of β's, τ's, power and correlation parameters.

Author(s)

Wagner Hugo Bonat

`mc_id` *Independent Model Structure*

Description

Builds an identity matrix to be used as a component of the matrix linear predictor. It is in general the first component of the matrix linear predictor, a kind of intercept matrix.

Usage

```
mc_id(data)
```

Arguments

`data`	the data set to be used.

Value

A list of matrix.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`mc_dist`, `mc_ma`, `mc_rw` and `mc_mixed`.

`mc_initial_values` *Automatic Initial Values*

Description

This function provides initial values to be used when fitting multivariate covariance generalized linear models by using the function `mcglm`. In general the users do not need to use this function, since it is already employed when setting the argument `control_initial = "automatic"` in the `mcglm` function. However, if the users want to change some of the initial values, this function can be useful.

Usage

```
mc_initial_values(linear_pred, matrix_pred, link, variance,
                  covariance, offset, Ntrial, contrasts, data)
```

Arguments

`linear_pred`	a list of formula see `formula` for details.
`matrix_pred`	a list of known matrices to be used on the matrix linear predictor. See `mc_matrix_linear_predictor` for details.
`link`	a list of link functions names, see `mcglm` for details.
`variance`	a list of variance functions names, see `mcglm` for details.
`covariance`	a list of covariance link functions names, see `mcglm` for details.
`offset`	a list of offset values if any.
`Ntrial`	a list of the number of trials on Bernoulli experiments. It is useful only for `"binomialP"` and `"binomialPQ"` variance functions.
`contrasts`	list of contrasts to be used in the `model.matrix`.
`data`	data frame.

Details

To obtain initial values for multivariate covariance generalized linear models the function `mc_initial_values` fits a generalized linear model (GLM) using the function `glm` with the specified linear predictor and link function for each response variables considering independent observations. The `family` argument is always specified as `quasi`. The link function depends on the specification of the argument `link`. The variance function is always specified as `"mu"` the only excession appears when using `variance = "constant"` then the family argument in the `glm` function is specified as `quasi(link = link, variance = "constant")`. The estimated value of the dispersion parameter from the `glm` function is used as initial value for the first component of the matrix linear predictor, for all other components the value zero is used.

For the cases `covariance = "inverse"` and `covariance = "expm"` the inverse and the logarithm of the estimated dispersion parameter is used as initial value for the first component of the matrix linear predictor. The value of the power parameter is always started at 1. In the cases of multivariate models the correlation between response variables is always started at 0.

Value

Return a list of initial values to be used while fitting in the `mcglm` function.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

`mc_link_function` *Link Functions*

Description

The `mc_link_function` is a customized call of the `make.link` function.

Given the name of a link function, it returns a list with two elements. The first element is the inverse of the link function applied on the linear predictor $\mu = g^{-1}(X\beta)$. The second element is the derivative of μ with respect to the regression parameters β. It will be useful when computing the quasi-score function.

Usage

```
mc_link_function(beta, X, offset, link)

mc_logit(beta, X, offset)

mc_probit(beta, X, offset)

mc_cauchit(beta, X, offset)

mc_cloglog(beta, X, offset)

mc_loglog(beta, X, offset)

mc_identity(beta, X, offset)

mc_log(beta, X, offset)

mc_sqrt(beta, X, offset)

mc_invmu2(beta, X, offset)

mc_inverse(beta, X, offset)
```

Arguments

- `beta` — a numeric vector of regression parameters.
- `X` — a design matrix, see `model.matrix` for details.
- `offset` — a numeric vector of offset values. It will be sum up on the linear predictor as a covariate with known regression parameter equals one ($\mu = g^{-1}(X\beta + offset)$). If no offset is present in the model, set offset = NULL.
- `link` — a string specifying the name of the link function. Options are: `"logit"`, `"probit"`, `"cauchit"`, `"cloglog"`, `"loglog"`, `"identity"`, `"log"`, `"sqrt"`, `"1/mu^2"` and `inverse`. A user defined link function can be used (see Details).

Details

The link function is an important component of the multivariate covariance generalized linear models, since it links the expectation of the response variable with the covariates. Let β be a (p x 1) regression parameter vector and X be an (n x p) design matrix. The expected value of the response variable Y is given by

$$E(Y) = g^{-1}(X\beta),$$

where g is the link function and $\eta = X\beta$ is the linear predictor. Let D be a (n x p) matrix whose entries are given by the derivatives of μ with respect to β. Such a matrix will be required for the fitting algorithm. The function `mc_link_function` returns a list where the first element is μ (n x 1) vector and the second is the D (n x p) matrix. A user defined function can also be used. It must be a function with arguments `beta`, `X` and `offset` (set to `NULL` if non needed). The function must return a length 2 named list with `mu` and `D` elements as a vector and a matrix of proper dimensions.

Value

A list with two elements: mu and D (see Details).

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

See Also

`model.matrix`, `make.link`.

Examples

```
x1 <- seq(-1, 1, l = 5)
X <- model.matrix(~ x1)
mc_link_function(beta = c(1,0.5), X = X,
                 offset = NULL, link = 'log')
mc_link_function(beta = c(1,0.5), X = X,
                 offset = rep(10,5), link = 'identity')
```

`mc_list2vec` *Auxiliar function transforms list to a vector.*

Description

This function takes a list of parameters and tranforms to a vector.

Usage

```
mc_list2vec(list_initial, list_power_fixed)
```

Arguments

`list_initial` A list specifying initial values.

`list_power_fixed` A list of logical operators specyfing if the power parameter should be estimated or not.

Details

It is an internal function, in general the users never will use this function. It will be useful, only if the user wants to implement a different variance-covariance matrix.

Value

A vector of model parameters.

Author(s)

Wagner Hugo Bonat

`mc_ma` *Moving Average Models Structure*

Description

The function `mc_ma` helps to build the components of the matrix linear predictor associated with moving average models. This function is generaly used for the analysis of longitudinal and times series data. The user can specify the order of the moving average process.

Usage

```
mc_ma(id, time, data, order = 1)
```

Arguments

`id` name of the column (string) containing the subject index. Note that this structure was designed to deal with longitudinal data. For times series data use the same `id` for all observations (one unit sample).

`time` name of the column (string) containing the index indicating the time.

`data` data set.

`order` order of the moving average process.

Details

This function was designed mainly to deal with longitudinal data, but can also be used for times series analysis. In that case, the `id` argument should contain only one index. It pretends a longitudinal data taken just for one individual or unit sample. This function is a simple call of the `bandSparse` function from the `Matrix` package.

Value

A matrix of `dgCMatrix` class.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`mc_id`, `mc_dist`, `mc_car`, `mc_rw` and `mc_mixed`.

Examples

```
id <- rep(1:2, each = 4)
time <- rep(1:4, 2)
data <- data.frame("id" = id, "time" = time)
mc_ma(id = "id", time = "time", data = data, order = 1)
mc_ma(id = "id", time = "time", data = data, order = 2)
```

---

mc_manova *MANOVA-type test for mcglm objects.*

Description

IT IS AN EXPERIMENTAL FUNCTION BE CAREFUL! MANOVA-type test for multivariate covariance generalized linear models.

Usage

```
mc_manova(object, ...)
```

Arguments

`object`	an object of `mcglm` class.
`...`	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

MANOVA table for mcglm objects.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

`mc_manova_disp` *MANOVA-type test for mcglm objects.*

Description

IT IS AN EXPERIMENTAL FUNCTION BE CAREFUL! MANOVA-type test for multivariate covariance generalized linear models. This function is specify for the dispersion components.

Usage

```
mc_manova_disp(object, idx, names, ...)
```

Arguments

`object`	an object of `mcglm` class.
`idx`	vector of indexes for parameter tests.
`names`	vector of names to appear in the ANOVA table.
`...`	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

MANOVA table for mcglm objects.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

`mc_matrix_linear_predictor` *Matrix Linear Predictor*

Description

Compute the matrix linear predictor. It is an internal function, however, since the concept of matrix linear predictor was proposed recently. I decided let this function visible to the interested reader gets some feeling about how it works.

Usage

```
mc_matrix_linear_predictor(tau, Z)
```

Arguments

`tau`	a numeric vector of dispersion parameters.
`Z`	a list of known matrices.

Details

Given a list with a set of known matrices $(Z_0, ..., Z_D)$ the function
`mc_matrix_linear_predictor` returns $U = \tau_0 Z_0 + ... + \tau_D Z_D$.

Value

A matrix.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

See Also

`mc_id`, `mc_dist`, `mc_ma`, `mc_rw`, `mc_mixed` and `mc_car`.

Examples

```
require(Matrix)
Z0 <- Diagonal(5, 1)
Z1 <- Matrix(rep(1,5)%*%t(rep(1,5)))
Z <- list(Z0, Z1)
mc_matrix_linear_predictor(tau = c(1,0.8), Z = Z)
```

`mc_mixed` *Mixed Models Structure*

Description

The function `mc_mixed` helps to build the components of the matrix linear predictor associated with mixed models. It is useful to model the covariance structure as a function of known covariates in a linear mixed model fashion (Bonat, et. al. 2016). The `mc_mixed` function was designed to analyse repeated measures and longitudinal data, where in general the observations are taken at a fixed number of groups, subjects or unit samples.

Usage

```
mc_mixed(formula, data)
```

Arguments

`formula`	a formula model to build the matrix linear predictor. See details.
`data`	data set.

Details

The `formula` argument should be specified similar to the linear predictor for the mean structure, however the first component should be 0 and the second component should always indicate the name of the column containing the subject or unit sample index. It should be a `factor`. The other covariates are specified after a slash "\" in the usual way. For example, `~0 + SUBJECT/(x1 + x2)` means that the column SUBJECT contains the subject or unit sample index, while the covariates that can be continuous or factors are given in the columns x1 and x2. Be careful the parenthesis after the "\" are mandatory, when including more than one covariate. The special case where only the SUBJECT effect is requested the formula takes the form `~ 0 + SUBJECT` without any extra covariate. This structure corresponds to the well known compound symmetry structure. By default the function `mc_mixed` include all interaction terms, the users can ignore the interactions terms removing them from the matrix linear predictor.

Value

A list of matrices.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Bonat, et. al. (2016). Modelling the covariance structure in marginal multivariate count models: Hunting in Bioko Island. Journal of Agricultural Biological and Environmental Statistics, 22(4):446–464.

See Also

`mc_id`, `mc_conditional_test`, `mc_dist`, `mc_ma`, `mc_rw` and `mc_car`.

Examples

```
SUBJECT <- gl(2, 6)
x1 <- rep(1:6, 2)
x2 <- rep(gl(2,3),2)
data <- data.frame(SUBJECT, x1 , x2)
# Compound symmetry structure
mc_mixed(~0 + SUBJECT, data = data)
# Compound symmetry + random slope for x1 and interaction or correlation
mc_mixed(~0 + SUBJECT/x1, data = data)
# Compound symmetry + random slope for x1 and x2 plus interactions
mc_mixed(~0 + SUBJECT/(x1 + x2), data = data)
```

mc_ns *Non-structure Model Structure*

Description

The function `mc_non` builds the components of the matrix linear predictor used for fitting non-structured covariance matrix. In general this model is hard to fit due to the large number of parameters.

Usage

```
mc_ns(id, data, group = NULL, marca = NULL)
```

Arguments

`id`	name of the column (string) containing the subject index. Note this structure was designed to deal with longitudinal data. For times series or spatial data use the same id for all observations (one unit sample).
`data`	data set.
`group`	name of the column (string) containing a group specific for which the covariance should change.
`marca`	level (string) of the column group for which the covariance should change.

Value

A list of a n*(n-1)/2 matrices.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`mc_id`, `mc_dglm`, `mc_dist`, `mc_ma`, `mc_rw`
and `mc_mixed`.

mc_pearson *Pearson estimating function*

Description

Compute the Pearson estimating function its sensitivity and variability matrices.

Usage

```
mc_pearson(
  y_vec,
  mu_vec,
  Cfeatures,
  inv_J_beta = NULL,
  D = NULL,
  correct = FALSE,
  compute_sensitivity = TRUE,
  compute_variability = FALSE,
  W
)
```

Arguments

`y_vec`	A vector.
`mu_vec`	A vector.
`Cfeatures`	A list of matrices.
`inv_J_beta`	A matrix.
`D`	A matrix.
`correct`	Logical.
`compute_sensitivity`	Logical.
`compute_variability`	Logical.
`W`	Matrix of weights.

Details

Compute the Pearson estimating function its sensitivity and variability matrices. For more details see Bonat and Jorgensen (2016) equations 6, 7 and 8.

Value

A list with three components: (i) a vector of quasi-score values, (ii) the sensitivity and (iii) variability matrices associated with the Pearson estimating function.

Author(s)

Wagner Hugo Bonat

mc_quasi_score *Quasi-score function*

Description

Compute the quasi-score function, its sensitivy and variability matrix.

Usage

```
mc_quasi_score(D, inv_C, y_vec, mu_vec, W)
```

Arguments

D	A matrix. In general the output from mc_link_function.
inv_C	A matrix. In general the output from mc_build_C.
y_vec	A vector.
mu_vec	A vector.
W	Matrix of weights.

Value

The quasi-score vector, the Sensivity and variability matrices.

Author(s)

Wagner Hugo Bonat

mc_remove_na *Remove NA from Matrix Linear Predictor*

Description

The function mc_remove_na removes NA from each component of the matrix linear predictor. It is in general used after the function mc_complete_data.

Usage

```
mc_remove_na(matrix_pred, cod)
```

Arguments

matrix_pred	a list of known matrices.
cod	index indicating the columns should be removed.

Value

A list of matrices.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`mc_dglm`, `mc_ns`, `mc_ma` and `mc_rw`.

`mc_robust_std` *Robust Standard Error for Regression Parameters*

Description

Compute robust standard error for regression parameters in the context of clustered observations for an object of `mcglm` class.

Usage

```
mc_robust_std(object, id)
```

Arguments

`object`	an object of `mcglm` class.
`id`	a vector which identifies the clusters or subject indexes. The length and order of `id` should be the same as the number of observations. Data are assumed to be sorted so that observations on a cluster are contiguous rows for all entities in the formula.

Value

A variance-covariance matrix. Note that the function assumes that the data are in the correct order.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Nuamah, I. F. and Qu, Y. and Aminu, S. B. (1996). A SAS macro for stepwise correlated binary regression. Computer Methods and Programs in Biomedicine 49, 199–210.

See Also

`mc_bias_correct_std`.

mc_rw *Random Walk Models Structure*

Description

The function mc_rw builds the components of the matrix linear predictor associated with random walk models. This function is generaly used for the analysis of longitudinal and times series data. The user can specify the order of the random walk process.

Usage

```
mc_rw(id, time, data, order = 1, proper = FALSE)
```

Arguments

id	name of the column (string) containing the subject index. Note that this structure was designed to deal with longitudinal data. For times series data use the same id for all observations (one unit sample).
time	name of the column (string) containing the index indicating the time.
data	data set.
order	order of the random walk model.
proper	logical.

Value

If proper = FALSE a matrix of dgCMatrix class. If proper = TRUE a list with two matrices of dgCMatrix class.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_id, mc_dist, mc_car, mc_ma, mc_mixed and mc_compute_rho.

Examples

```
id <- rep(1:2, each = 4)
time <- rep(1:4, 2)
data <- data.frame("id" = id, "time" = time)
mc_rw(id = "id", time = "time", data = data, order = 1, proper = FALSE)
mc_rw(id = "id", time = "time", data = data, order = 1, proper = TRUE)
mc_rw(id = "id", time = "time", data = data, order = 2, proper = TRUE)
```

`mc_sandwich` *Matrix product in sandwich form*

Description

The function `mc_sandwich` is just an auxiliar function to compute product matrix in the sandwich form `bord1 * middle * bord2`. An special case appears when computing the derivative of the covariance matrix with respect to the power parameter. Always the bord1 and bord2 should be diagonal matrix. If it is not true, this product is too slow.

Usage

```
mc_sandwich(middle, bord1, bord2)

mc_sandwich_negative(middle, bord1, bord2)

mc_sandwich_power(middle, bord1, bord2)

mc_sandwich_cholesky(bord1, middle, bord2)

mc_multiply(bord1, bord2)

mc_multiply2(bord1, bord2)
```

Arguments

`middle`	A matrix.
`bord1`	A matrix.
`bord2`	A matrix.

Value

The matrix product `bord1 * middle * bord2`.

Author(s)

Wagner Hugo Bonat

`mc_sensitivity` *Sensitivity matrix*

Description

Compute the sensitivity matrix associated with the Pearson estimating function.

Usage

```
mc_sensitivity(product, W)
```

Arguments

`product`	A list of matrix.
`W`	Matrix of weights.

Details

This function implements the equation 7 of Bonat and Jorgensen (2016).

Value

The sensitivity matrix associated with the Pearson estimating function.

Author(s)

Wagner Hugo Bonat and Eduardo Elias Ribeiro Jr

`mc_sic` *Score Information Criterion - Regression*

Description

Compute the score information criterion (SIC) for an object of `mcglm` class. The SIC is useful for selecting the components of the linear predictor. It can be used to construct an stepwise covariate selection.

Usage

```
mc_sic(object, scope, data, response, penalty = 2, weights)
```

Arguments

`object`	an object of `mcglm` class.
`scope`	a vector of covariate names to be tested.
`data`	data set containing all variables involved in the model.
`response`	index indicating for which response variable the SIC should be computed.
`penalty`	penalty term (default = 2).
`weights`	Vector of weights for model fitting.

Value

A data frame containing SIC values, degree of freedom, Tu-statistics and chi-squared reference values.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Bonat, et. al. (2016). Modelling the covariance structure in marginal multivariate count models: Hunting in Bioko Island. Journal of Agricultural Biological and Environmental Statistics, 22(4):446–464.

See Also

mc_sic_covariance.

Examples

```
set.seed(123)
x1 <- runif(100, -1, 1)
x2 <- gl(2,50)
beta = c(5, 0, 3)
X <- model.matrix(~ x1 + x2)
y <- rnorm(100, mean = X%*%beta , sd = 1)
data <- data.frame(y, x1, x2)
# Reference model
Z0 <- mc_id(data)
fit0 <- mcglm(linear_pred = c(y ~ 1), matrix_pred = list(Z0), data = data)
# Computing SIC
mc_sic(fit0, scope = c("x1","x2"), data = data, response = 1)
```

---

mc_sic_covariance *Score Information Criterion - Covariance*

Description

Compute the score information criterion (SIC) for an object of mcglm class. The SIC-covariance is useful for selecting the components of the matrix linear predictor. It can be used to construct an stepwise procedure to select the components of the matrix linear predictor.

Usage

```
mc_sic_covariance(object, scope, idx, data, penalty = 2, response, weights)
```

Arguments

object	an object of mcglm class.
scope	a list of matrices to be tested.
idx	indicator of matrices belong to the same effect. It is useful for the case where more than one matrix represents the same effect.
data	data set containing all variables involved in the model.
penalty	penalty term (default = 2).
response	index indicating for which response variable SIC-covariance should be computed.
weights	Vector of weights for model fitting.

Value

A data frame containing SIC-covariance values, degree of freedom, Tu-statistics and chi-squared reference values for each matrix in the scope argument.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, et. al. (2016). Modelling the covariance structure in marginal multivariate count models: Hunting in Bioko Island. Journal of Agricultural Biological and Environmental Statistics, 22(4):446–464.

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_sic.

Examples

```
set.seed(123)
SUBJECT <- gl(10, 10)
y <- rnorm(100)
data <- data.frame(y, SUBJECT)
Z0 <- mc_id(data)
Z1 <- mc_mixed(~0+SUBJECT, data = data)
# Reference model
fit0 <- mcglm(c(y ~ 1), list(Z0), data = data)
# Testing the effect of the matrix Z1
mc_sic_covariance(fit0, scope = Z1, idx = 1,
data = data, response = 1)
# As expected Tu < Chisq indicating non-significance of Z1 matrix
```

---

mc_transform_list_bdiag

Auxiliar function to compute the derivatives of the C matrix.

Description

This function take a list of matrices and return a list of block-diagonal matrices, where the original matrices are one block non-zero of the matrix.

Usage

```
mc_transform_list_bdiag(list_mat, mat_zero, response_number)
```

Arguments

`list_mat` A list of matrices.

`mat_zero` A list of zero matrices. In general the output of `link[mcglm]{mc_build_bdiag}`.

`response_number` A numeric specifying the response variable number.

Value

A list of block-diagonal matrices.

Author(s)

Wagner Hugo Bonat

`mc_twin` *Twin Models Structure*

Description

The function `mc_twin` helps to build the components of the matrix linear predictor associated with ACDE models for analysis of twin data.

Usage

```
mc_twin(id, twin.id, type, replicate = NULL, structure, data)

mc_twin_bio(id, twin.id, type, replicate = NULL, structure, data)

mc_twin_full(id, twin.id, type, replicate, formula, data)
```

Arguments

`id` name of the column (string) containing the twin index. It should be the same index (number) for both twins.

`twin.id` name of the column (string) containing the twin index inside the pair. In general 1 for the first twin and 2 for the second twin.

`type` name of the column (string) containing the indication of the twin as mz or dz. It should be a factor with only two levels mz and dz. Be sure that the reference level is mz.

`replicate` name of the column (string) containing the index for more than one observation taken at the same twin pair. It is used for example in twin longitudinal studies. In that case, the replication column should contain the time index.

`structure` model type options are full, flex, uns, ACE, ADE, AE, CE and E. See example for details.

`data` data set.

`formula` internal.

Value

A list of matrices of `dgCMatrix` class.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

mc_id, mc_dist, mc_car, mc_rw, mc_ns, mc_dglm and mc_mixed.

Examples

```
id <- rep(1:5, each = 4)
id.twin <- rep(1:2, 10)
```

---

mc_updateBeta *Updated regression parameters*

Description

This function update a list of regression parameters. It will be useful only inside the fitting algorithm.

Usage

```
mc_updateBeta(list_initial, betas, information, n_resp)
```

Arguments

list_initial	A list of initial values.
betas	A vector with actual regression parameters values.
information	A list with information about the number of parameters in the model. In general the output from mc_getInformation.
n_resp	A numeric specyfing the number of response variables.

Value

A list with updated values of the regression parameters.

Author(s)

Wagner Hugo Bonat

mc_updateCov — *Updated covariance parameters*

Description

This function update a list of covariance parameters. It will be useful only inside the fitting algorithm.

Usage

```
mc_updateCov(list_initial, covariance, list_power_fixed, information, n_resp)
```

Arguments

`list_initial`	A list of initial values.
`covariance`	A vector with actual covariance parameters values.
`list_power_fixed`	A list of logicals indicating if the power parameter should be estimated or not.
`information`	A list with information about the number of parameters in the model. In general the output from mc_getInformation.
`n_resp`	A numeric specyfing the number of response variables.

Value

A list with updated values of the covariance parameters.

Author(s)

Wagner Hugo Bonat

mc_variability — *Variability matrix*

Description

Compute the variability matrix associated with the Pearson estimating function.

Usage

```
mc_variability(sensitivity, product, inv_C, C, res, W)
```

Arguments

`sensitivity`	A matrix. In general the output from `mc_sensitivity`.
`product`	A list of matrix.
`inv_C`	A matrix. In general the output from `mc_build_C`.
`C`	A matrix. In general the output from `mc_build_C`.
`res`	A vector. The residuals vector, i.e. (y_vec - mu_vec).
`W`	Matrix of weights.

Details

This function implements the equation 8 of Bonat and Jorgensen (2016).

Value

The variability matrix associated witht the Pearson estimating function.

Author(s)

Wagner Hugo Bonat and Eduardo Elias Ribeiro Jr

---

mc_variance_function *Variance Functions*

Description

Compute the variance function and its derivatives with respect to regression, dispersion and power parameters.

Usage

```
mc_variance_function(mu, power, Ntrial, variance, inverse,
                          derivative_power, derivative_mu)

mc_power(mu, power, inverse, derivative_power, derivative_mu)

mc_binomialP(mu, power, inverse, Ntrial,
                    derivative_power, derivative_mu)

mc_binomialPQ(mu, power, inverse, Ntrial,
                    derivative_power, derivative_mu)
```

Arguments

mu	a numeric vector. In general the output from mc_link_function.
power	a numeric value (power and binomialP) or a vector (binomialPQ) of the power parameters.
Ntrial	number of trials, useful only when dealing with binomial response variables.
variance	a string specifying the name (power, binomialP or binomialPQ) of the variance function.
inverse	logical. Compute the inverse or not.
derivative_power	logical if compute (TRUE) or not (FALSE) the derivatives with respect to the power parameter.
derivative_mu	logical if compute (TRUE) or not (FALSE) the derivative with respect to the mu parameter.

Details

The function mc_variance_function computes three features related with the variance function. Depending on the logical arguments, the function returns $V^{1/2}$ and its derivatives with respect to the parameters power and mu, respectivelly. The output is a named list, completely informative about what the function has been computed. For example, if inverse = FALSE, derivative_power = TRUE and derivative_mu = TRUE. The output will be a list, with three elements: V_sqrt, D_V_sqrt_power and D_V_sqrt_mu.

Value

A list with from one to four elements depends on the arguments.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

See Also

mc_link_function.

Examples

```
x1 <- seq(-1, 1, l = 5)
X <- model.matrix(~x1)
mu <- mc_link_function(beta = c(1, 0.5), X = X, offset = NULL,
                       link = "logit")
mc_variance_function(mu = mu$mu, power = c(2, 1), Ntrial = 1,
                     variance = "binomialPQ", inverse = FALSE,
                     derivative_power = TRUE, derivative_mu = TRUE)
```

NewBorn *Respiratory Physiotherapy on Premature Newborns.*

Description

The NewBorn dataset consists of a prospective study to assess the effect of respiratory physiotherapy on the cardiopulmonary function of ventilated preterm newborn infants with birth weight lower than 1500 g. The data set was collected and kindly made available by the nursing team of the Waldemar Monastier hospital, Campo Largo, PR, Brazil. The NewBorn dataset was analysed in Bonat and Jorgensen (2016) as an example of mixed outcomes regression model.

- Sex - Factor two levels Female and Male.
- GA - Gestational age (weeks).
- BW - Birth weight (mm).
- APGAR1M - APGAR index in the first minute of life.

- APGAR5M - APGAR index in the fifth minute of life.
- PRE - Factor, two levels (Premature: YES; NO).
- HD - Factor, two levels (Hansen's disease, YES; NO).
- SUR - Factor, two levels (Surfactant, YES; NO).
- JAU - Factor, two levels (Jaundice, YES; NO).
- PNE - Factor, two levels (Pneumonia, YES; NO).
- PDA - Factor, two levels (Persistence of ductus arteriosus, YES; NO).
- PPI - Factor, two levels (Primary pulmonary infection, YES; NO).
- OTHERS - Factor, two levels (Other diseases, YES; NO).
- DAYS - Age (days).
- AUX - Factor, two levels (Type of respiratory auxiliary, HOOD; OTHERS).
- RR - Respiratory rate (continuous).
- HR - Heart rate (continuous).
- SPO2 - Oxygen saturation (bounded).
- TREAT - Factor, three levels (Respiratory physiotherapy, Evaluation 1; Evaluation 2; Evaluation 3).
- NBI - Newborn index.
- TIME - Days of treatment.

Usage

```
data(NewBorn)
```

Format

a data.frame with 270 records and 21 variables.

Source

Bonat, W. H. and Jorgensen, B. (2016) Multivariate covariance generalized linear models. Journal of Royal Statistical Society - Series C 65:649–675.

Examples

```
library(mcglm)
library(Matrix)
data(NewBorn, package="mcglm")
formu <- SPO2 ~ Sex + APGAR1M + APGAR5M + PRE + HD + SUR
Z0 <- mc_id(NewBorn)
fit <- mcglm(linear_pred = c(formu), matrix_pred = list(Z0),
            link = c("logit"), variance = c("binomialP"),
            power_fixed = c(TRUE),
            data = NewBorn,
            control_algorithm = list(verbose = FALSE, tuning = 0.5))
summary(fit)
```

pAIC — *Pseudo Akaike Information Criterion*

Description

Extract the pseudo Akaike information criterion (pAIC) for objects of `mcglm` class.

Usage

```
pAIC(object, verbose = TRUE)
```

Arguments

`object`	an object or a list of objects representing a model of `mcglm` class.
`verbose`	logical. Print or not the pAIC value.

Value

Returns the value of the pseudo Akaike information criterion (pAIC).

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`gof`, `plogLik`, `ESS`, `pKLIC`, `GOSHO` and `RJC`.

pBIC — *Pseudo Bayesian Information Criterion*

Description

Extract the pseudo Bayesian information criterion (pBIC) for objects of `mcglm` class.

Usage

```
pBIC(object, verbose = TRUE)
```

Arguments

`object`	an object or a list of objects representing a model of `mcglm` class.
`verbose`	logical. Print or not the pBIC value.

Value

Returns the value of the pseudo Bayesian information criterion (pBIC).

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`gof`, `plogLik`, `ESS`, `pKLIC`, `GOSHO` and `RJC`.

---

pKLIC — *Pseudo Kullback-Leibler Information Criterion*

Description

Extract the pseudo Kullback-Leibler information criterion (pKLIC) for objects of `mcglm` class.

Usage

```
pKLIC(object, verbose = TRUE)
```

Arguments

`object`	an object or a list of objects representing a model of `mcglm` class.
`verbose`	logical. Print or not the pKLIC value.

Value

Returns the value of the pseudo Kullback-Leibler information criterion.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

See Also

`gof`, `plogLik`, `ESS`, `pAIC`, `GOSHO` and `RJC`.

plogLik	*Gaussian Pseudo-loglikelihood*

Description

Extract the Gaussian pseudo-loglikelihood (plogLik) value for objects of `mcglm` class.

Usage

```
plogLik(object, verbose = TRUE)
```

Arguments

`object`	an object or a list of objects representing a model of `mcglm` class.
`verbose`	logical. Print or not the plogLik value.

Value

Returns the value of the Gaussian pseudo-loglikelihood.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

plot.mcglm	*Residuals and algorithm check plots*

Description

Residual and algorithm check analysis for objects of `mcglm` class.

Usage

```
## S3 method for class 'mcglm'
plot(x, type = "residuals", ...)
```

Arguments

`x`	a fitted `mcglm` object.
`type`	specify which graphical analysis will be performed. Options are: `"residuals"` and `"algorithm"`.
`...`	additional arguments affecting the plot produced. Note that there is no extra options for mcglm object class.

Value

The function `plot.mcglm` was designed to offer a fast residuals analysis based on the Pearson residuals. Current version offers a simple Pearson residuals versus fitted values and a quantile plot. When using `algorithm = TRUE` the function will plot a summary of the fitting algorithm shows the trajectory or iterations of the fitting algorithm. The iterations are shown in terms of values of the model parameters and also the actually value of the quasi-score and Pearson estimating functions. Hence, a quickly check of the algorithm convergence is obtained.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

See Also

`residuals` and `fitted`.

`print.mcglm` *Print*

Description

The default print method for an object of `mcglm` class.

Usage

```
## S3 method for class 'mcglm'
print(x, ...)
```

Arguments

`x`	fitted model objects of class `mcglm` as produced by `mcglm()`.
`...`	further arguments passed to or from other methods.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

See Also

`summary`.

`residuals.mcglm` *Residuals*

Description

Compute residuals for an object of `mcglm` class.

Usage

```
## S3 method for class 'mcglm'
residuals(object, type = "raw", ...)
```

Arguments

`object`	an object of `mcglm` class.
`type`	the type of residuals which should be returned. Options are: `"raw"` (default), `"pearson"` and `"standardized"`.
`...`	additional arguments affecting the residuals produced. Note that there is no extra options for mcglm object class.

Value

The function `residuals.mcglm` returns a matrix of residuals values.

Author(s)

Wagner Hugo Bonat, `<wbonat@ufpr.br>`

See Also

`fitted`.

`RJC` *Rotnitzky-Jewell Information Criterion*

Description

Compute the Rotnitzky-Jewell information criterion for an object of `mcglm` class. WARNINGS: This function is limited to models with ONE response variable.

Usage

```
RJC(object, id, verbose = TRUE)
```

Arguments

`object`	an object of `mcglm` class.
`id`	a vector which identifies the clusters. The length and order of `id` should be the same as the number of observations. Data are assumed to be sorted so that observations on a cluster are contiguous rows for all entities in the formula.
`verbose`	logical. Print or not the RJC value.

Value

The value of the Rotnitzky-Jewell information criterion. Note that the function assumes that the data are in the correct order.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

Source

Wang, M. (2014). Generalized Estimating Equations in Longitudinal Data Analysis: A Review and Recent Developments. Advances in Statistics, 1(1)1–13.

See Also

`gof`, `plogLik`, `pAIC`, `pKLIC`, `ESS` and `GOSHO`.

`soil` *Soil Chemistry Properties Data*

Description

Soil chemistry properties measured on a regular grid with 10 x 25 points spaced by 5 meters.

- `COORD.X` - X coordinate.
- `COORD.Y` - Y coordinate.
- `SAND` - Sand portion of the sample.
- `SILT` - Silt portion of the sample.
- `CLAY` - Clay portion of the sample.
- `PHWATER` - Soil pH at water.
- `CA` - Calcium content.
- `MG` - Magnesium content.
- `K` - Potassio content.

Usage

```
data(soil)
```

Format

a `data.frame` with 250 records and 9 variables.

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Examples

```
data(soil, package="mcglm")
## neigh <- spdep::tri2nb(soil[,1:2])
## Spatial model
## Z1 <- mc_car(neigh) take too long
Z1 <- mc_id(soil)
# Linear predictor
form.ca <- CA ~ COORD.X*COORD.Y + SAND + SILT + CLAY + PHWATER
fit.ca <- mcglm(linear_pred = c(form.ca), matrix_pred = list(Z1),
               link = "log", variance = "tweedie", covariance = "inverse",
               power_fixed = TRUE, data = soil,
               control_algorith = list(max_iter = 1000, tuning = 0.1,
               verbose = FALSE, tol = 1e-03))
## mc_compute_rho(fit.ca) only for spatial model
```

---

soya *Soybeans*

Description

Experiment carried out in a vegetation house with soybeans. The experiment has two plants by plot with three levels of the factor amount of water in the soil (`water`) and five levels of potassium fertilization (`pot`). The plots were arranged in five blocks (`block`). Three response variables are of the interest, namely, grain yield, number of seeds and number of viable peas per plant. The data set has 75 observations of 7 variables.

- `pot` - Factor five levels of potassium fertilization.
- `water` - Factor three levels of amount of water in the soil.
- `block` - Factor five levels.
- `grain` - Continuous - Grain yield per plant.
- `seeds` - Count - Number of seeds per plant.
- `viablepeas` - Binomial - Number of viable peas per plant.
- `totalpeas` - Binomial - Total number of peas per plant.

Usage

```
data(soya)
```

Format

a `data.frame` with 75 records and 7 variables.

Source

Bonat, W. H. (2018). Multiple Response Variables Regression Models in R: The mcglm Package. Journal of Statistical Software, 84(4):1–30.

Examples

```
library(mcglm)
library(Matrix)
data(soya, package="mcglm")
formu <- grain ~ block + factor(water) * factor(pot)
Z0 <- mc_id(soya)
fit <- mcglm(linear_pred = c(formu), matrix_pred = list(Z0),
          data = soya)
          anova(fit)
```

---

summary.mcglm *Summarizing*

Description

The default summary method for an object of `mcglm` class.

Usage

```
## S3 method for class 'mcglm'
summary(
  object,
  verbose = TRUE,
  print = c("Regression", "power", "Dispersion", "Correlation"),
  ...
)
```

Arguments

`object`	an object of `mcglm` class.
`verbose`	logical. Print or not the model summary.
`print`	print only part of the model summary, options are `Regression`, `power`, `Dispersion` and `Correlation`.
`...`	additional arguments affecting the summary produced. Note the there is no extra options for mcglm object class.

Value

Print a `mcglm` object.

Author(s)

Wagner Hugo Bonat, <`wbonat@ufpr.br`>

See Also

`print`.

vcov.mcglm *Variance-Covariance Matrix*

Description

Returns the variance-covariance matrix for an object of mcglm class.

Usage

```
## S3 method for class 'mcglm'
vcov(object, ...)
```

Arguments

object	an object of mcglm class.
...	additional arguments affecting the summary produced. Note that there is no extra options for mcglm object class.

Value

A variance-covariance matrix.

Author(s)

Wagner Hugo Bonat, <wbonat@ufpr.br>

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